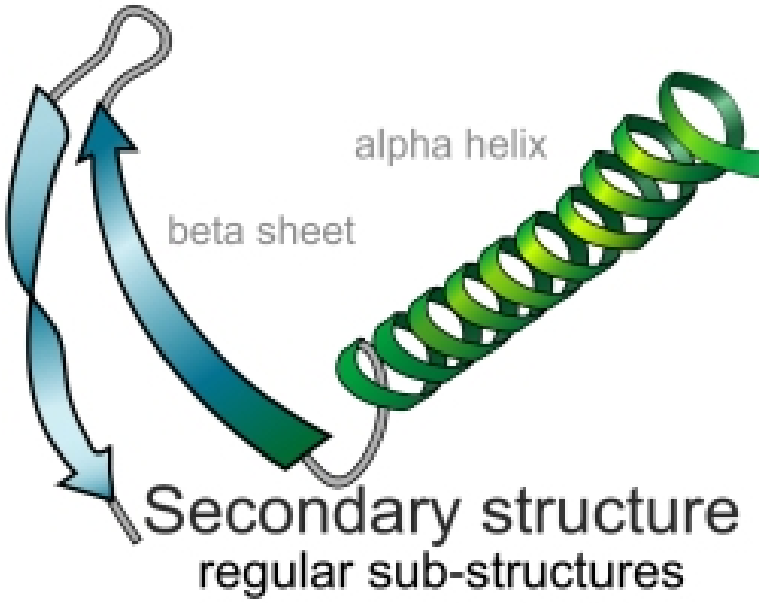
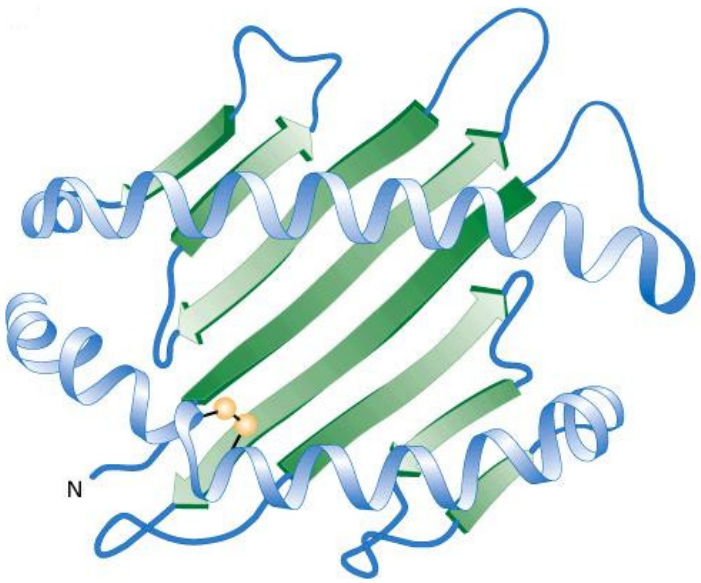
Lecture 9: Neural Networks for Biological Prediction

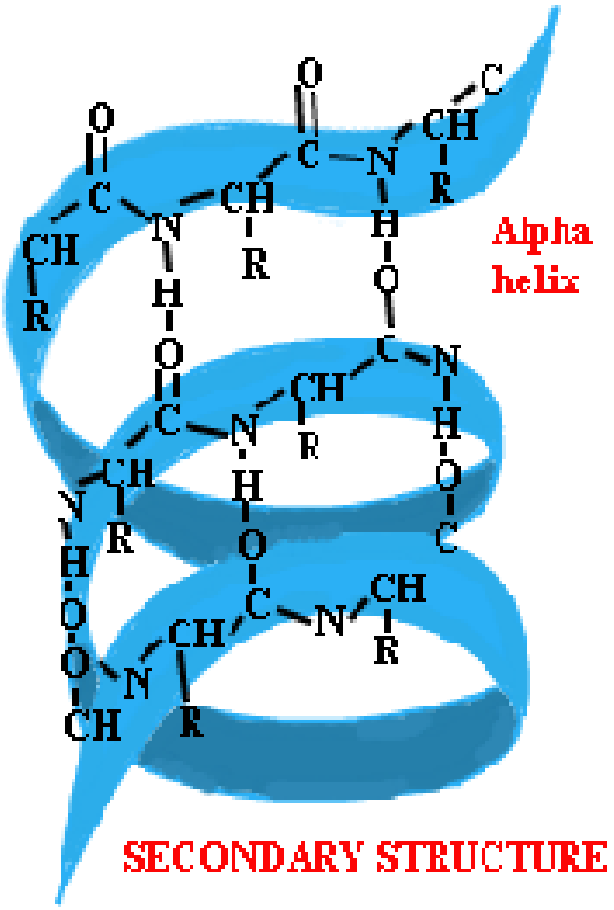




MLP, One-Hot Vectors, Softmax, Profile, Psi-Blast

Outline

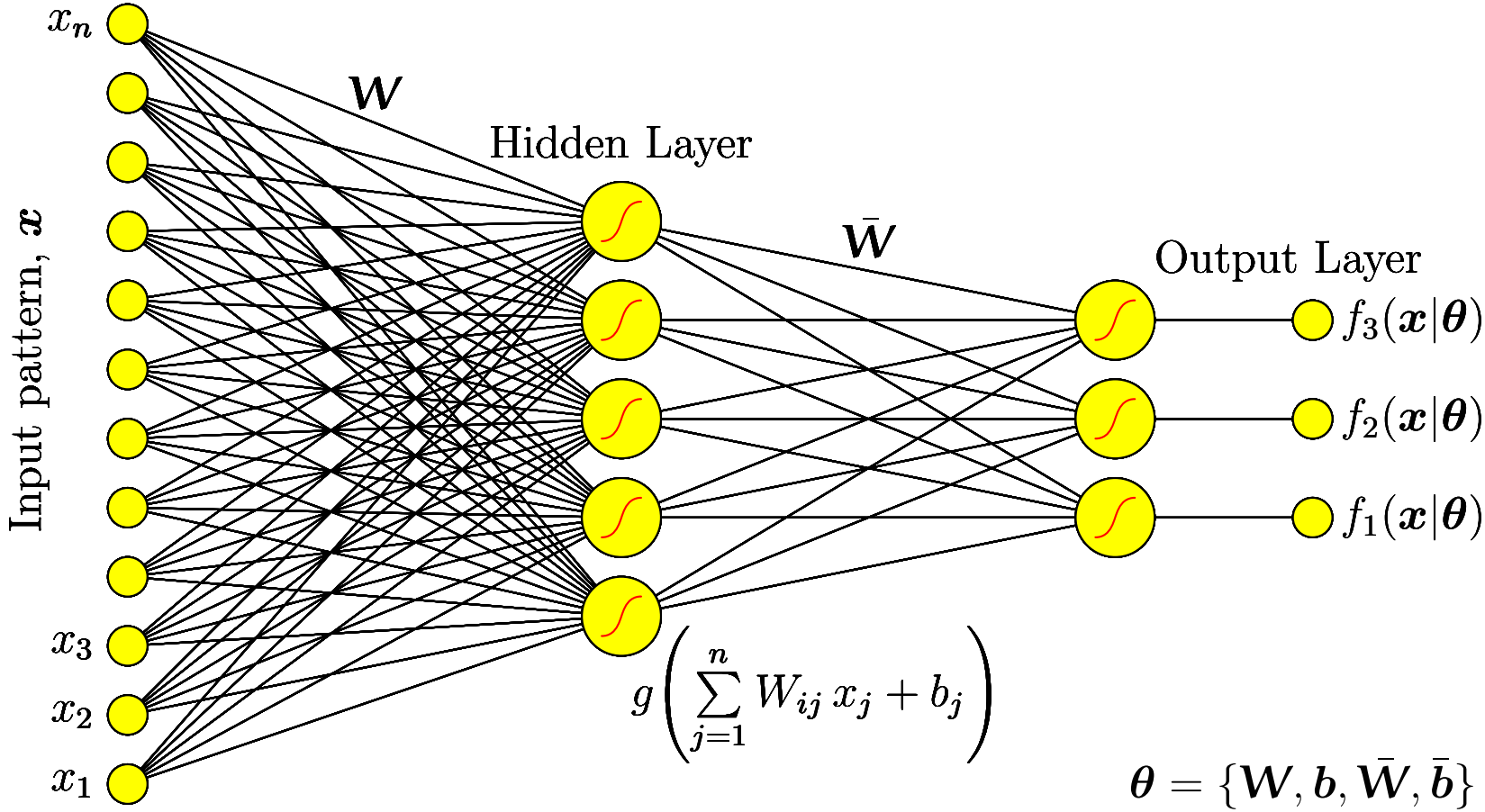
1. **MLP**
2. Secondary Protein Structure
3. Designing MLPs



Neural Networks in Biology

- Neural networks (and support vector machines) are frequently used to make prediction in biology
- We are not going into detail about neural networks (many of you have done this in machine learning)
- However, how to use them in biology is quite subtle and is the main topic of this lecture
- We take as a case study the classic problem of the prediction of secondary structure of proteins
- Although, some of this is dated, the lessons are timeless

Multilayer Perceptron (MLP)



$$\boldsymbol{\theta} = \{\boldsymbol{W}, \boldsymbol{b}, \bar{\boldsymbol{W}}, \bar{\boldsymbol{b}}\}$$

$$f_i(\boldsymbol{x}|\boldsymbol{\theta}) = g\left(\sum_{j=1}^{n_h} \bar{W}_{i,j}\, g\left(\sum_{k=1}^{n} W_{ik}\, x_k + b_k\right) + \bar{b}_i\right)$$

Training Neural Networks

- We train a neural network on a set of data

$$\mathcal{D} = \left((\boldsymbol{x}^\mu, \boldsymbol{y}^\mu) \,\middle|\, \mu = 1,\, 2,\, \ldots,\, m \right)$$

- Define an loss function $L(\boldsymbol{\theta}, \mathcal{D})$

- E.g. squared error

$$L(\boldsymbol{\theta}, \mathcal{D}) = \frac{1}{m} \sum_{\mu=1}^{m} \left\| \boldsymbol{y}^\mu - \boldsymbol{f}(\boldsymbol{x}^\mu | \boldsymbol{\theta}) \right\|^2$$

Gradient Descent

- To minimise the loss we follow the negative gradient

$$\boldsymbol{\nabla} L(\boldsymbol{\theta}, \mathcal{D}) = \begin{pmatrix} \frac{\partial L(\boldsymbol{\theta},\mathcal{D})}{\partial \theta_1} \\ \frac{\partial L(\boldsymbol{\theta},\mathcal{D})}{\partial \theta_2} \\ \vdots \\ \frac{\partial L(\boldsymbol{\theta},\mathcal{D})}{\partial \theta_N} \end{pmatrix}$$

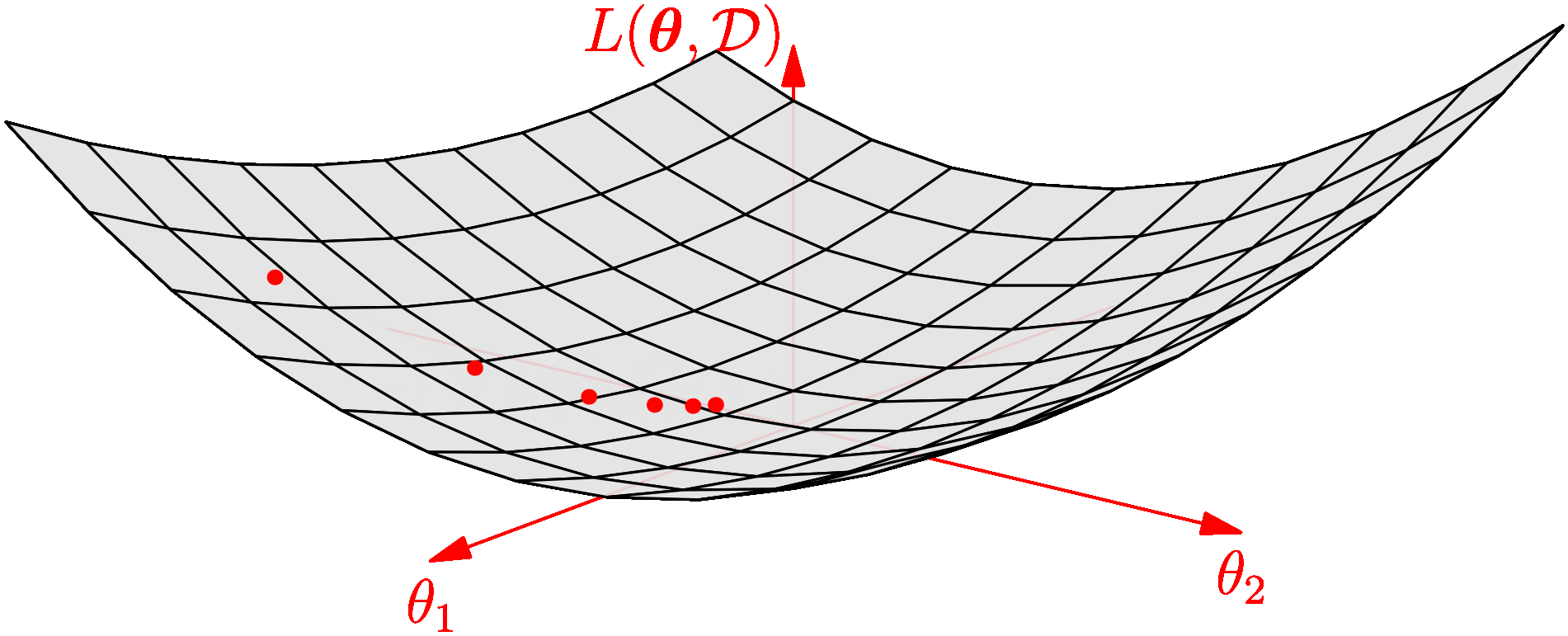
- where $\theta_1 = W_{11}$, $\theta_2 = W_{12}$, etc.

- We make a step in the direction of the negative gradient

$$\boldsymbol{\theta}(t+1) = \boldsymbol{\theta}(t) - r\, \boldsymbol{\nabla} L(\boldsymbol{\theta}, \mathcal{D})$$

- r is learning rate

Following the Gradient

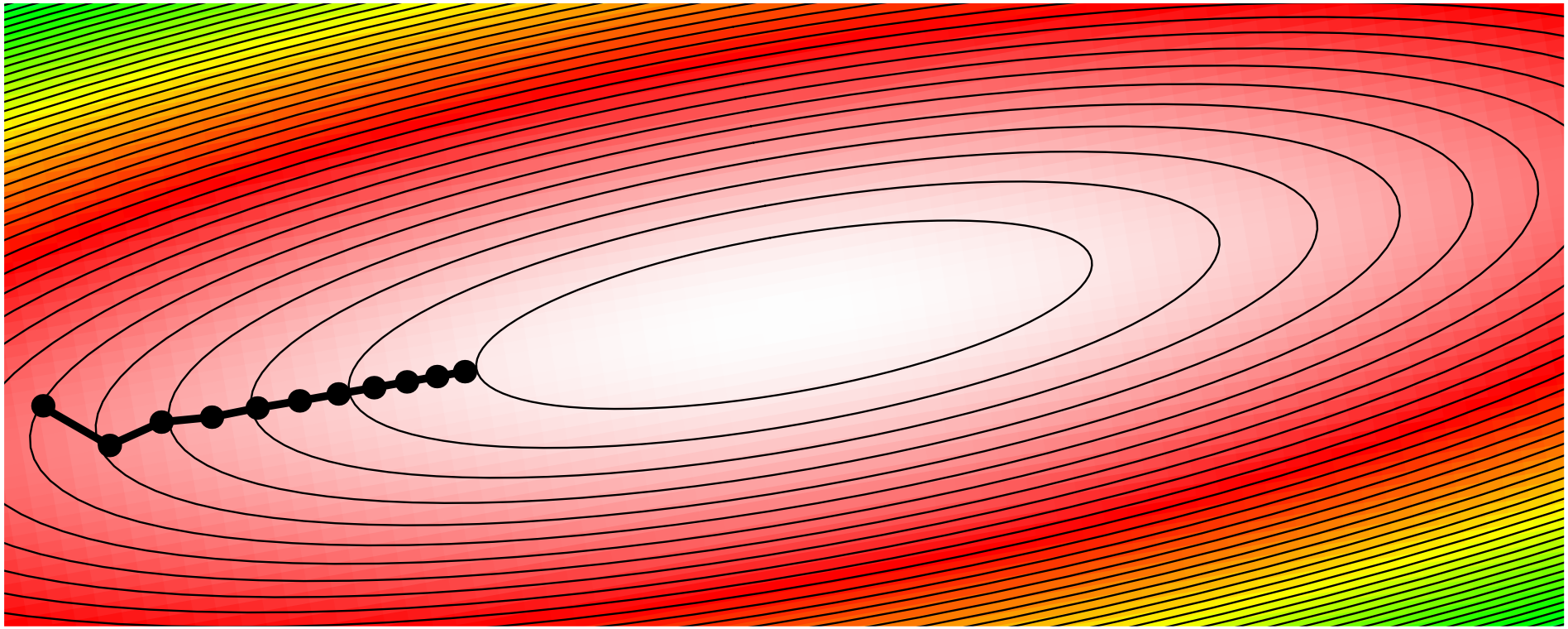


Backpropagation

- Calculating the gradient for a MLP is known as back-propagation
- We can compute the gradient of the error for the whole training set
- In this case we often use Levenberg-Marquardt, scale conjugate gradient or BFGS to speed up convergence
- More often it is effective to find the error per input pattern or mini-batch (a small number of input patterns)
- This goes by the name of stochastic gradient descent
- This introduces noise which helps escape local optima

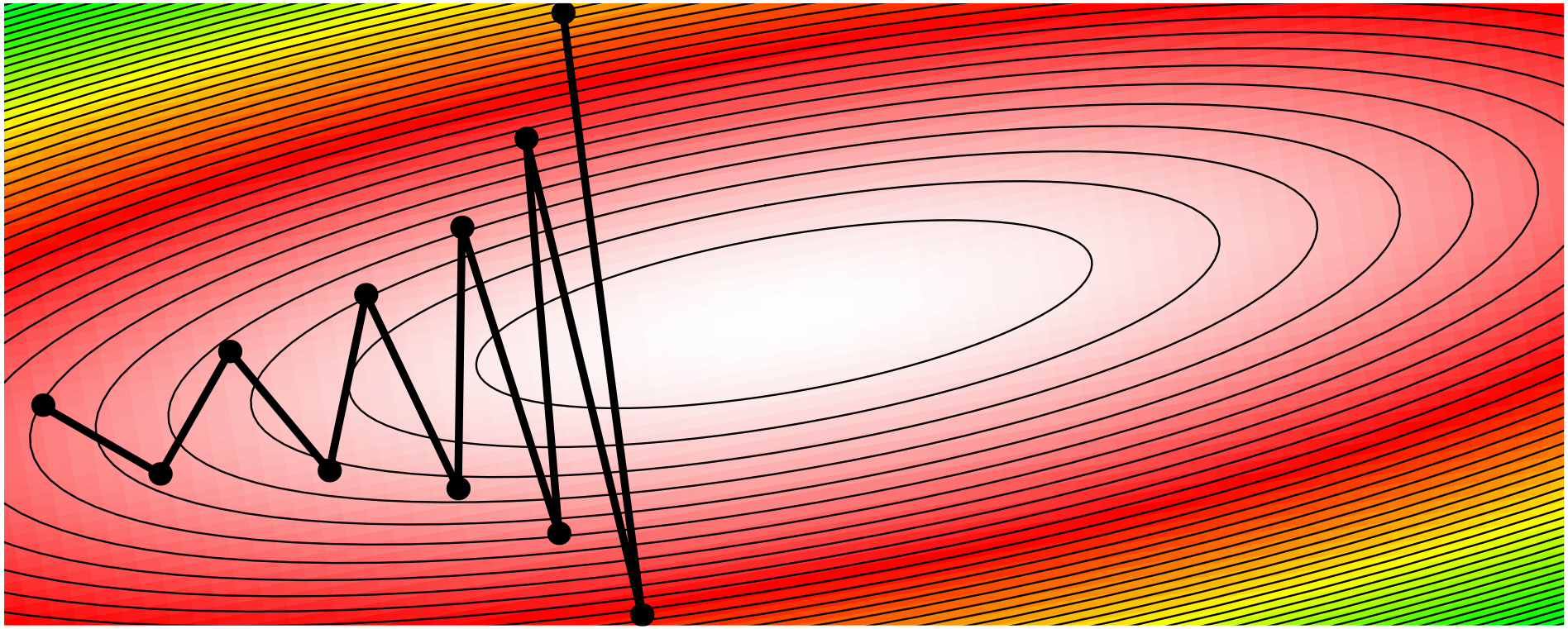
Higher Dimensions

- In higher dimensions the problem is that there are some directions you need to move a long way



More Haste Less Speed

- Increasing the step size a little increases the rate of converge in one direction, but . . .



Speeding Up Stochastic Gradient Descent

- To speed up stochastic gradient descent we can add a momentum term

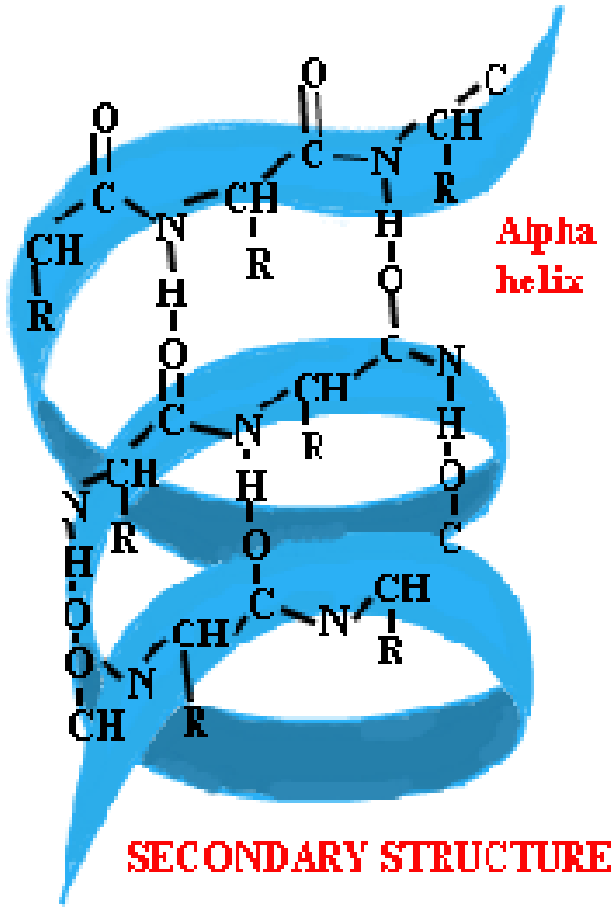
$$\boldsymbol{v}(t) = (1-\epsilon)\boldsymbol{v}(t) - r\boldsymbol{\nabla} L(\boldsymbol{\theta}(t-1))$$
$$\boldsymbol{\theta}(t) = \boldsymbol{\theta}(t-1) + \boldsymbol{v}(t)$$

- Many variations on this, e.g. ada-delta and adam

- These are quite new techniques developed for deep-learning, but equally applicable to MLPs

Outline

1. MLP
2. **Secondary Protein Structure**
3. Designing MLPs



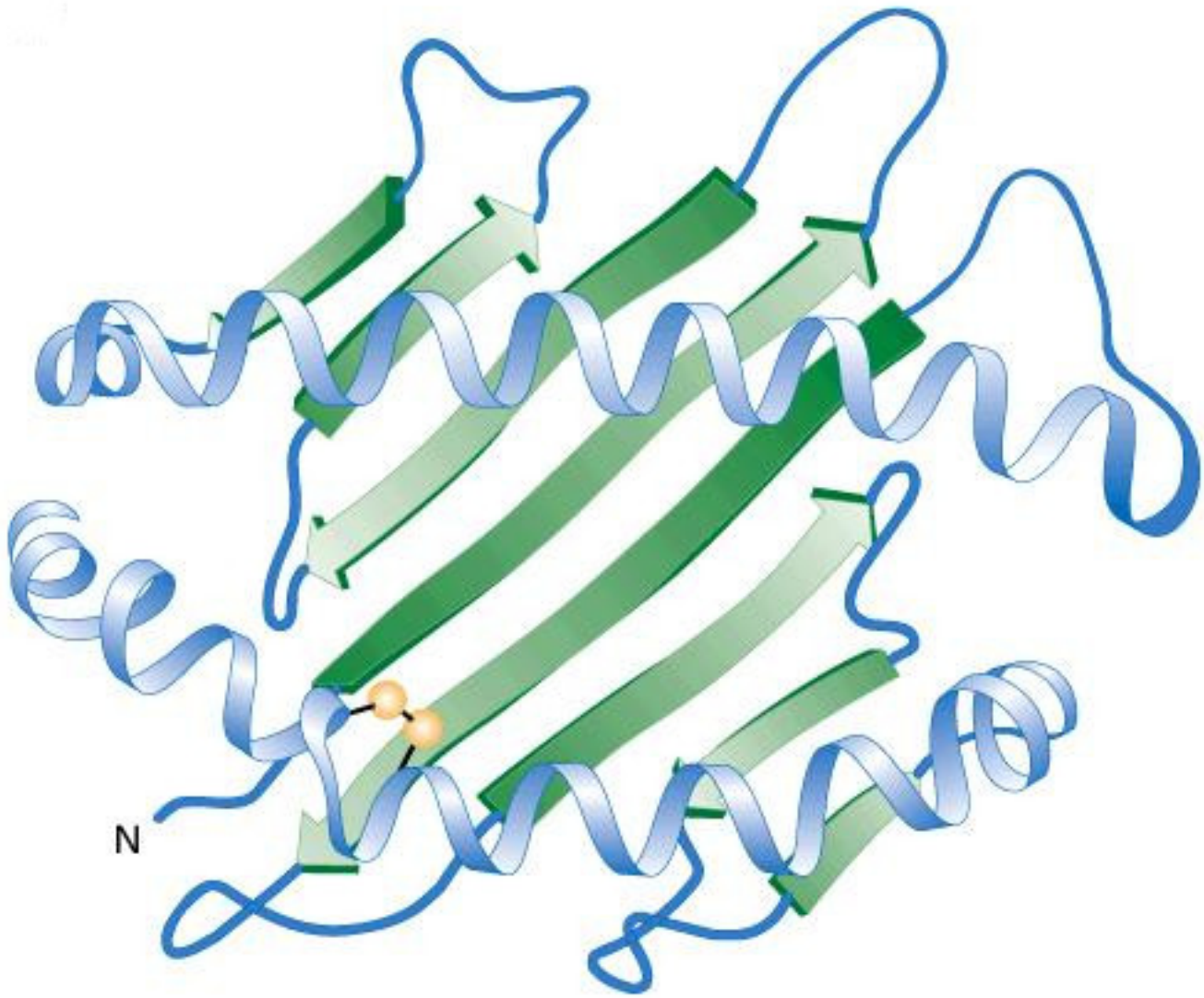
Protein Structure

- Understanding the 3D structure of proteins is very important to understand its function
- Unfortunately, it is extremely difficult to infer its 3D structure
  - ⋆ It is both experimentally very costly
  - ⋆ And computationally very challenging
- However knowing a proteins **primary** structure (sequence) is quite easy (we can read this off if we know the RNA sequence from which it is translated)

Secondary Structure

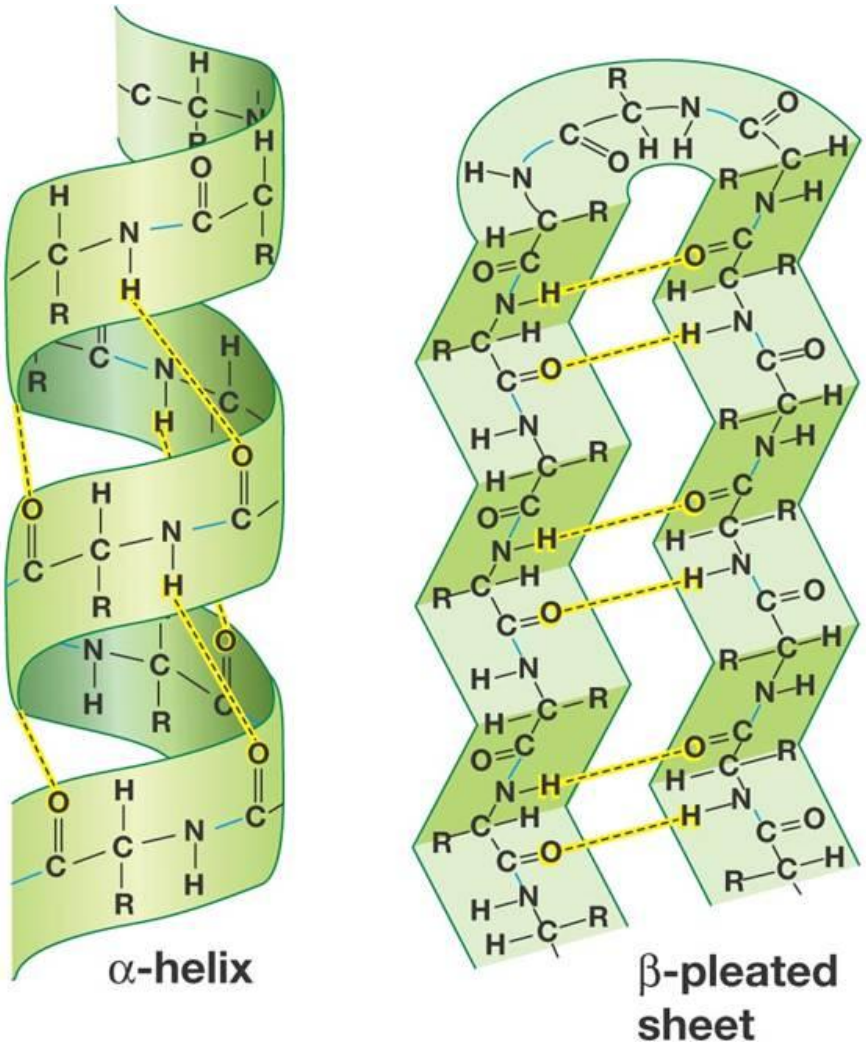
- The secondary structure tells us whether a protein is part of
  - ⋆ An alpha-helix
  - ⋆ A beta-sheet
  - ⋆ Something else (often referred to as coil)
- This does not tell us the full 3-D structure but it is the first step towards inferring this

Protein Structure



Test Problem

- Secondary protein structure has long been a classic test problem for machine learning
- It was an early success for MLPs
- People have tried SVMs, HMMs, etc.
- It is a difficult problem in part because it is non-local



Outline

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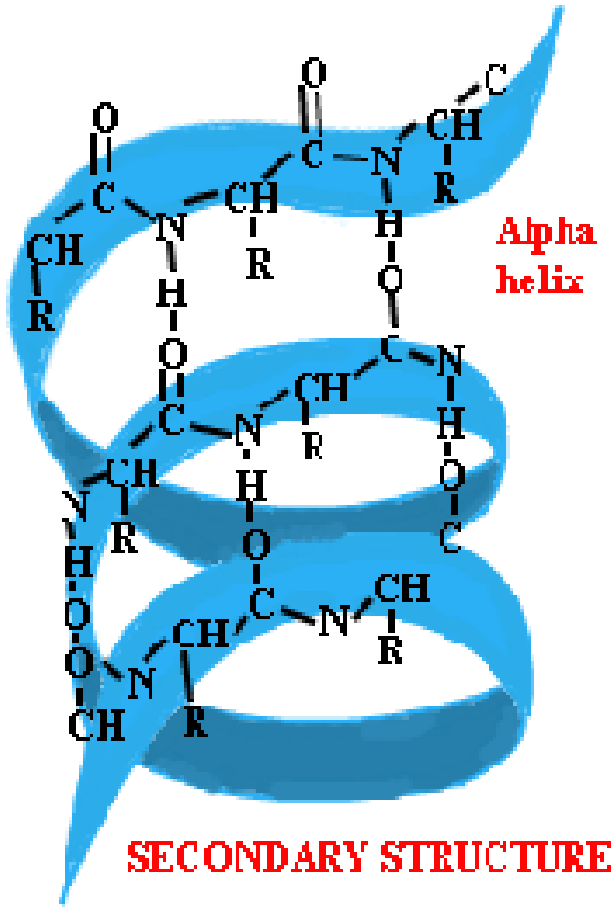
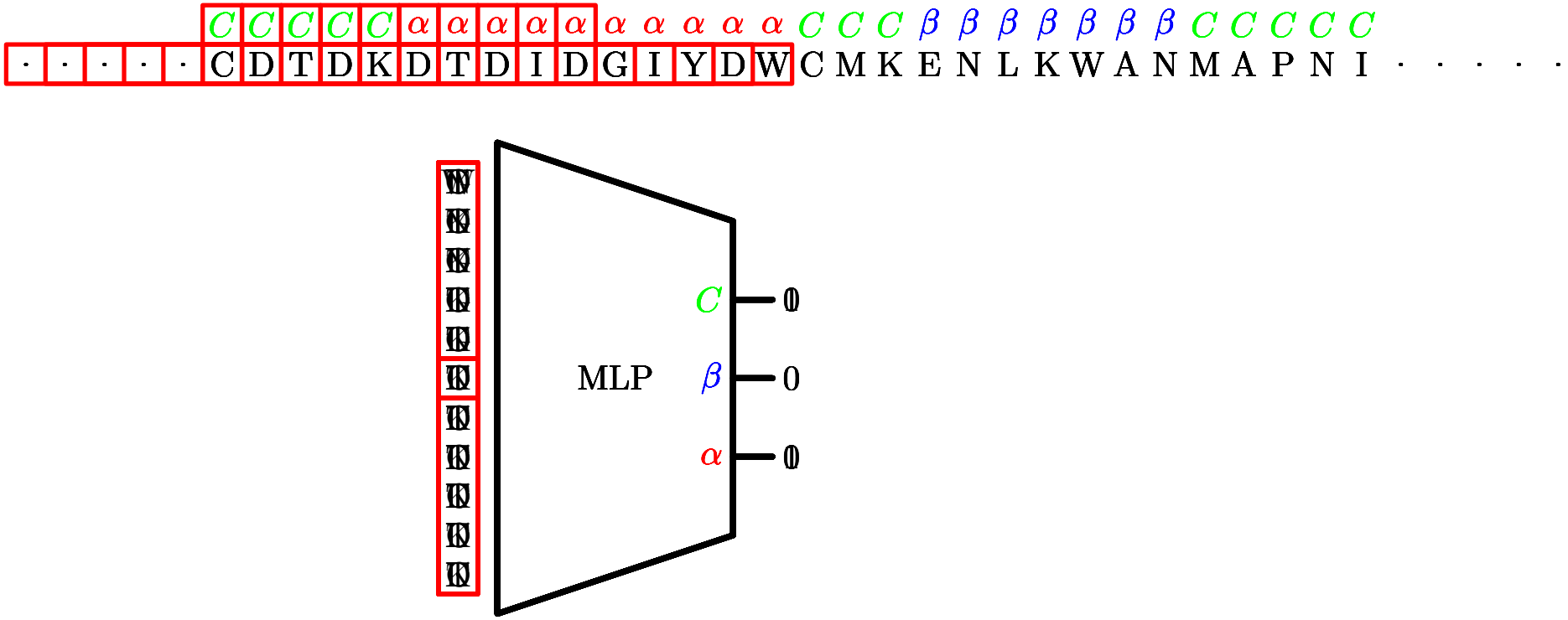
3. **Designing MLPs**



Predicting Secondary Structure

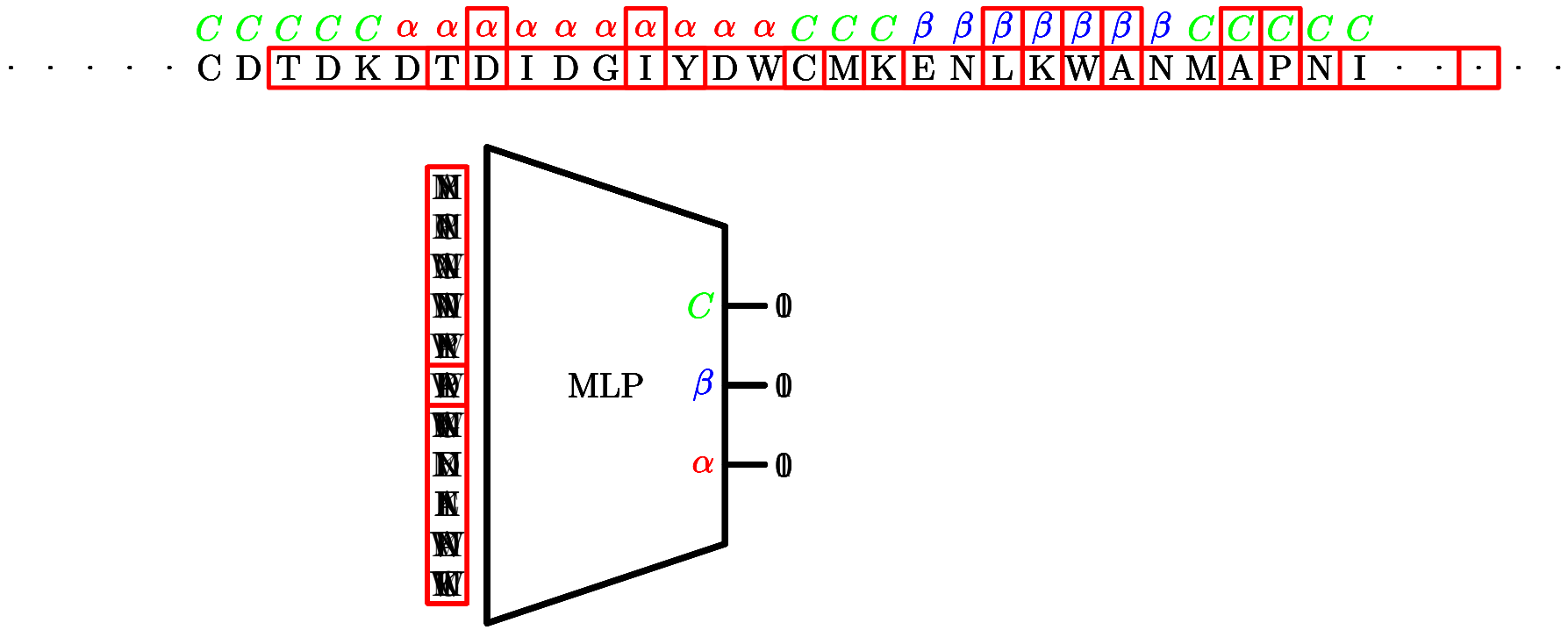
- Although very costly we know the secondary structure for quite a few proteins
- This provides a training and testing data set
- However, its not immediately obvious how to create a neural network to use this data
- Most learning machines take a fixed length input

Windowing


C C C C C α α α α α α α α α α C C C β β β β β β β C C C C C
· · · · · C D T D K D T D I D G I Y D W C M K E N L K W A N M A P N I · · · · ·
MLP
C
β
α
0
0
0

Randomised Windows

- During training we can randomise the placement of the window so we don't learn a long section of one structure



Representing Proteins

- How should we feed our protein to an MLP?
- Providing a number from 1–20 would put in an artificial ordering
- We could try and put in chemical properties, e.g. acid/base hydophobic/hydophilic, etc. but which ones
- It turned out best to use a **1-hot** (orthogonal) representation

$$\begin{aligned}
\text{Glycine (G)} &= (1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0) \\
\text{Alanine (A)} &= (0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0) \\
&\vdots \\
\text{End} &= (0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1)
\end{aligned}$$

- This is unbiased and seems to work best

Output

- Usually have 3 output (alpha-helix, beta-sheet and coil) although sometimes coil is split into 6 classes
- Again use a 1-hot representation
- Use winner-takes-all prediction (largest output)

Softmax

- A slightly better network is to use a **softmax** output where

$$f_i(\boldsymbol{x}|\boldsymbol{\theta}) = \frac{\mathrm{e}^{\sum_{j=1}^{h} \bar{W}_{ij} h_j(\boldsymbol{x}) + \bar{b}_i}}{\sum_{k=1}^{3} \mathrm{e}^{\sum_{j=1}^{h} \bar{W}_{kj} h_j(\boldsymbol{x}) + \bar{b}_i}}$$

- where $h_j(\boldsymbol{x})$ is the output of the jth hidden unit

- Note that we can the interpret the output as

$$f_i(\boldsymbol{x}|\boldsymbol{\theta}) = \mathbb{P}\left(\text{class } i|\boldsymbol{x},\, \boldsymbol{\theta}\right)$$

Relative Entropy

- Given a training set $\mathcal{D} = \big((\boldsymbol{x}^\mu, y^\mu) \,\big|\, \mu = 1, 2, \ldots, P\big)$ where $y^\mu \in \mathcal{S} = \{\alpha\text{-helix}, \beta\text{-sheet}, \text{Coil}\}$, the log-likelihood of the data is

$$L(\boldsymbol{\theta}) = \log(\mathbb{P}\,(\mathcal{D}|\boldsymbol{\theta})) = \sum_{\mu=1}^{P} \log(\mathbb{P}\,(y^\mu|\boldsymbol{x}^\mu, \boldsymbol{\theta}))$$

$$= \sum_{\mu=1}^{P} \log(f_{y^\mu}(\boldsymbol{x}^\mu|\boldsymbol{\theta}))$$

- This is known as the **cross entropy**

- Rather than minimising the squared error we can maximise the relative entropy—again we use a gradient based technique

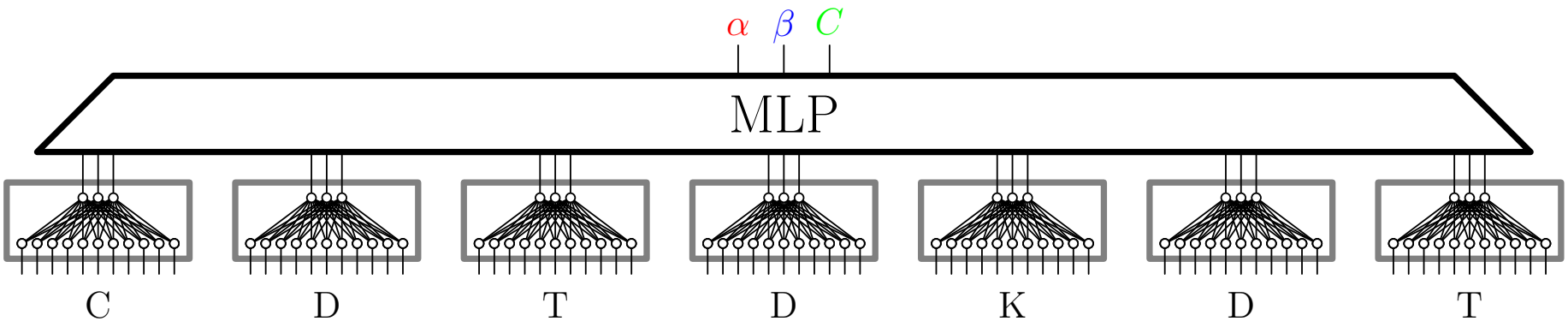
Simple ANN

- A simple network would consist of
  - $W \times 21$ inputs where W is the window size (typically around 11)
  - n_h hidden sigmoidal units (typically around 50)
  - 3 softmax output units
- Results are around 60% prediction of the three classes

Learning A Better Input Representation

- The 1-hot vector representation of the input makes the network very large

- We can learn an encoding or **embedding** by using the same 21 to M layer for all inputs in the window



- All the input encoding networks share the same weights

- This slightly complicates back-prop, but is not too difficult

Top Tier

- When testing we get a prediction at each site of the sequence
- We can improve our machine by building a second machine that takes the prediction at a site and its neighbours
- This uses slightly more global information and prevents silly predictions, e.g. α, β, α, C
- In machine learning this is known as stacking

Using Profiles

- The big improvement in prediction came from using the protein profile obtained by finding similar protein structures

```
A5ASC3.1    14 SIKLWPPSQTTRLLLVERMANNLST..PSIFTRK..YGSLSKEEARENAKQIEEVACSTANQ.....HYEKEPDGDGGSAVQLYAKECSKLILEVLK 101
B4F917.1    13 SIKLWPPSESTRIMLVDRMTNNLST..ESIFSRK..YRLLGKQEAHENAKTIEELCFALADE.....HFREEPDGDGSSAVQLYAKETSKMMLEVLK 100
A9S1V2.1    23 VFKLWPPSQGTREAVRQKMALKLSS..ACFESQS..FARIELADAQEHARAIEEVAFGAAQE......ADSGGDKTGSAVVMVYAKHASKLMLETLR 109
B9GSN7.1    13 SVKLWPPGQSTRLMLVERMTKNFIT..PSFISRK..YGLLSKEEAEEDAKKIEEVAFAAANQ.....HYEKQPDGDGSSAVQIYAKESSRLMLEVLK 100
Q8H056.1    30 SFSIWPPTQRTRDAVVRRLVDTLGG..DTILCKR..YGAVPAADAEPAARGIEAEAFDAAAA..SGEAAATASVEEGIKALQLYSKEVSRRLLDFVK 120
Q0D4Z3.2    44 SLSIWPPSQRTRDAVVRRLVQTLVA..PSILSQR..YGAVPEAEAGRAAAAVEAEAYAAVTES.SSAAAAPASVEDGIEVLQAYSKEVSRRLLELAK 135
B9MVW8.1    56 SFSIWPPTQRTRDAIISRLIETLST..TSVLSKR..YGTIPKEEASEASRRIEEEAFSGAST.......VASSEKDGLEVLQLYSKEISKRMLETVK 141
Q0IYC5.1    29 SFAVWPPTRRTRDAVVRRLVAVLSGDTTTALRKRYRYGAVPAADAERAARAVEAQAFDAASA....SSSSSSSVEDGIETLQLYSREVSNRLLAFVR 121
A9NW46.1    13 SIKLWPPSESTRLMLVERMTDNLSS..VSFFSRK..YGLLSKEEAAENAKRIEETAFLAAND.....HEAKEPNLDDSSVVQFYAREASKLMLEALK 100
Q9C500.1    57 SLRIWPPTQKTRDAVLNRLIETLST..ESILSKR..YGTLKSDDATTVAKLIEEEAYGVASN.......AVSSDDDGIKILELYSKEISKRMLESVK 142
Q2HRI7.1    25 MYSIWPPKQRTRDAVKNRLIETLST..PSVLTKR..YGTMSADEASAAAIQIEDEAFSVANA.......SSSTSNDNVTILEVYSKEISKRMIETVK 110
Q9M7N3.1    28 SFKIWPPTQRTREAVVRRLVETLTS..QSVLSKR..YGVIPEEDATSAARIIEEEAFSVASV.ASAASTGGRPEDEWIEVLHIYSQEIXQRVVESAK 119
Q9M7N6.1    25 SFSIWPPTQRTRDAVINRLIESLST..PSILSKR..YGTLPQDEASETARLIEEEAFAAAGS.......TASDADDGIEILQVYSKEISKRMIDTVK 110
Q9LE82.1    14 SVKMWPPSKSTRLMLVERMTKNITT..PSIFSRK..YGLLSVEEAEQDAKRIEDLAFATANK.....HFQNEPDGDGTSAVHVYAKESSKLMLDVIK 101
Q9M651.2    13 SIKLWPPSLPTRKALIERITNNFSS..KTIFTEK..YGSLTKDQATENAKRIEDIAFSTANQ.....QFEREPDGDGGSAVQLYAKECSKLILEVLK 100
B9R748.1    48 SLSIWPPTQRTRDAVITRLIETLSS..PSVLSKR..YGTISHDEAESAARRIEDEAFGVANT.......ATSAEDDGLEILQLYSKEISRRMLDTVK 133
```

Profile Encoding

- Instead of using the protein sequence as input we use a coding showing all possible substitutions, deletions or insertions at a site
- This works because protein structure is much better conserved than protein sequences
- Thus related protein sequences will with high probability have the same secondary structure
- The protein profiles are "smoothed" by the related sequence information
- This improved performance by $\sim 5\%$

PSIPRED

- Profiling can be taken further. Rather than just finding matching sequences we can iterate the process
- We use the matched sequences to find other related sequences
- This is what Psi-Blast does
- It slow, but it produces the best profiles
- The best predictor for many years was PSIPRED, an MLP that uses Psi-Blast profiles, getting close to 80% performance—now deep learning (CNNs, etc.) dominate

Summary

- Common machine learning tools have proved very useful in biological sequence analysis
- However, great care must be taken to ensure that the information fed into them is meaningful
- Often these have to be combined with other tools (such as profilers) to get really good performance